

II B.E. (Computer Engineering)
Class Test II October 2023

3CERG1: Computer Organization and Architecture

Time : 70 Min.

Max. Marks 20

Note: Attempt as much as you can. Maximum marks awarded will not exceed 20.

- Q1** (a) If a 2s complement binary number is represented in 4 bits. State whether decimal number +8 can be represented in 4bits. If your answer is yes then write its binary equivalent and if your answer is no then give its reason. (6)
- (b) If decimal numbers -5 and -4 are added using 4 bit 2s complement form then what conclusion you can draw on its result.
- Q2** Perform binary multiplication of -7×6 using Booths algorithm (4)
- Q3** How will you represent binary number 1001.101 in IEEE 754 floating point representation. (4)
- Q4** A RAM memory is organized as 128K X 8. (i) Determine total number of bits and bytes in this RAM. (ii) number of address lines required (iii) number of chips, if it is organized using 16K X 4 DRAM chip. Explain your answer. (6)
- Q5** A computer with 1 MB of main memory; word size of 4 bytes; block size of 16 bytes; and cache size of 64 kB. (6)
- a. For the main memory addresses of A0110, and CBABE, give corresponding tag, cache line address, and word offsets for a direct-mapped cache.
- b. For the main memory addresses of A0110, and CBABE, give the corresponding tag and offset values for a fully-associative cache.
- c. For the main memory addresses of A0110, and CBABE, give corresponding tag, cache set, and offset values for a two-way set-associative cache.
- Q6** Draw and explain 4M X 4 DRAM organization. (4)